

Joanna X. Ju

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LinkedIn – www.linkedin.com/in/joannaxiaolinju | Portfolio (GitHub) – <https://github.com/joannaxju>

Summary

Computer Engineering student at UC San Diego focused on computer architecture, digital hardware, and embedded systems. Experience in processor design, FPGA development, embedded firmware, and avionics, with an upcoming NASA internship in high-performance and real-time flight computing.

Education

University of California, San Diego – La Jolla, CA

Fall 2025 – Present

Bachelor of Science in Computer Engineering

Expected Graduation: June 2028

- Cumulative GPA: 3.70 / 4.00 (UC San Diego & Pasadena City College, Pasadena, CA)

Projects

Custom 8-bit RISC Processor Design – SystemVerilog, Assembly, C++

- Designed and implemented a custom 8-bit RISC processor, defining the ISA, instruction formats, registers, and addressing modes.
- Implemented the processor datapath, ALU, control logic, and memory interfaces using synthesizable SystemVerilog.
- Developed assembly programs and an assembler to translate assembly instructions into machine code and resolve label offsets.
- Verified processor functionality through simulation and FPGA testing, using results to refine the ISA and implementation.

Experience

Incoming High Performance Computing & Embedded Systems Intern

September 2026 – December 2026

NASA Armstrong Flight Research Center, Edwards, CA

- Selected to support NASA's High-Performance Spaceflight Computing (HPSC) platform and Next-generation Research Flight Controls System (NuRFCS).
- Expected to contribute to real-time, high-performance embedded software development in C/C++ for safety-critical flight systems.
- Anticipated work includes multithreaded programming, hardware-in-the-loop testing, and embedded communication across Linux and real-time operating systems.

SPIS Mentor

August 2026 – September 2026

UCSD CSE Department, La Jolla, CA

- Mentor incoming computer science and engineering students in algorithmic problem-solving and programming.
- Guide students in designing, programming, and debugging Raspberry Pi robotics projects using Python.

Software/Avionics Team Member

October 2025 – Present

UCSD Rocket Propulsion Laboratory (RPL), La Jolla, CA

- Collaborate with the avionics team to maintain and validate embedded systems for student-built rockets.
- Analyze C++/Arduino firmware and hardware interfaces for ground station and flight computer architecture, sensor communication, and system behavior.
- Test and validate flight computer and ground station functionality using simulation, diagnosing hardware-software integration and communication issues.
- Prototype and integrate circuits connecting sensors and electronic components to microcontrollers.

Computer Science Tutor

August 2024 – June 2025

Pasadena City College, Pasadena, CA

Math Instructor

July 2024 – August 2026

Mathnasium, Temple City & San Diego, CA

Skills

Digital Hardware: SystemVerilog, Verilog, RTL Design, FPGA, Computer Architecture, Processor Design

Embedded Systems: C/C++, Microcontrollers, Arduino, Sensor Integration, Ethernet, SPI, I2C

Programming: Python, Java, Assembly

Hardware & EDA: Cadence Virtuoso, Quartus, ModelSim/Questa, PSpice, LTspice, Oscilloscope, Multimeter

Tools: Git, GitHub, Linux, VS Code

Languages: English (Fluent), Mandarin Chinese (Fluent)